

Am I Audible Now? Dissecting Prosody in Neural Audio Codecs

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Abstract

The functional organization of prosody within the Residual Vector Quantization (RVQ) codebooks of neural audio codecs remains an open question. Prior work relies on correlational probing, showing that pitch features can be decoded from frozen codebook embeddings. Decodability, however, does not imply causal necessity: a codebook may encode pitch information the decoder ignores, while another may be essential despite containing little probe-recoverable signal. We address this gap through causal output-side ablation: for each codebook, we zero or mean-replace its discrete tokens, re-decode the waveform, and measure degradation in fundamental frequency F_0 (CREPE), energy (frame-wise RMS), and intelligibility (STOI) against the unablated reconstruction. Experiments span 200 LibriSpeech test-clean utterances under clean, 15, 5, and 0 dB SNR with MUSAN noise across multiple neural audio codecs. Across different architectures, we find a consistent two-tier causal structure: a codec-specific subset of codebooks is reconstruction-critical, while the remaining intelligibility-preserving codebooks exhibit a stable pitch-contribution hierarchy that persists under additive noise. These findings provide new evidence for how neural audio codecs causally organize prosodic information.

1 Introduction

Neural audio codecs (Zeghidour et al., 2021) compress speech into a short sequence of discrete tokens per frame and increasingly serve as the tokenization layer beneath real-time speech and audio language models (Li et al., 2024; Borsos et al., 2023b), which guides decisions about codebook pruning, quantizer-dropout schedules, and bitrate allocation under real-world noise (Tseng and Harwath, 2025).

A split-RVQ codec (Juang and Gray, 1982; Zeghidour et al., 2021) such as Mimi (Défossez et al., 2024) encodes audio with one semantic codebook followed by a cascade of acoustic codebooks, but the architecture does not specify where prosody is represented, and discrete-unit speech language models have historically discarded prosodic information by design (Kharitonov et al., 2022). Recent probing studies (Vigneaux and Cole, 2026; Ballier et al., 2026) find pitch recoverable from all eight Mimi codebooks, disputing a clean semantic/acoustic split – but that evidence is correlational: probe recoverability does not show the decoder causally relies on a codebook during reconstruction (Elazar et al., 2021). It remains unclear which codebooks causally determine prosodic reconstruction, or whether that causal structure is stable under noise.

We answer these questions through causal codebook ablation, measuring each codebook’s contribution to pitch and energy under clean and noisy conditions.

Our contributions are:

- We present the first causal, output-side analysis of prosodic information in neural audio codecs.
- We identify a two-level codebook structure: one codebook dominates overall reconstruction, while the remaining codebooks exhibit a stable hierarchy in their contribution to pitch.
- We show that this ranking holds under distribution shift with three levels of additive noise, and becomes more stable after excluding the confounded codebook.

2 Related Work

Neural audio codecs trace back to multi-stage vector quantization for speech coding (Juang and Gray, 1982); the current generation of split-RVQ codecs is exemplified by SoundStream (Zeghidour et al., 2021), SpeechTokenizer (Zhang et al., 2024), and

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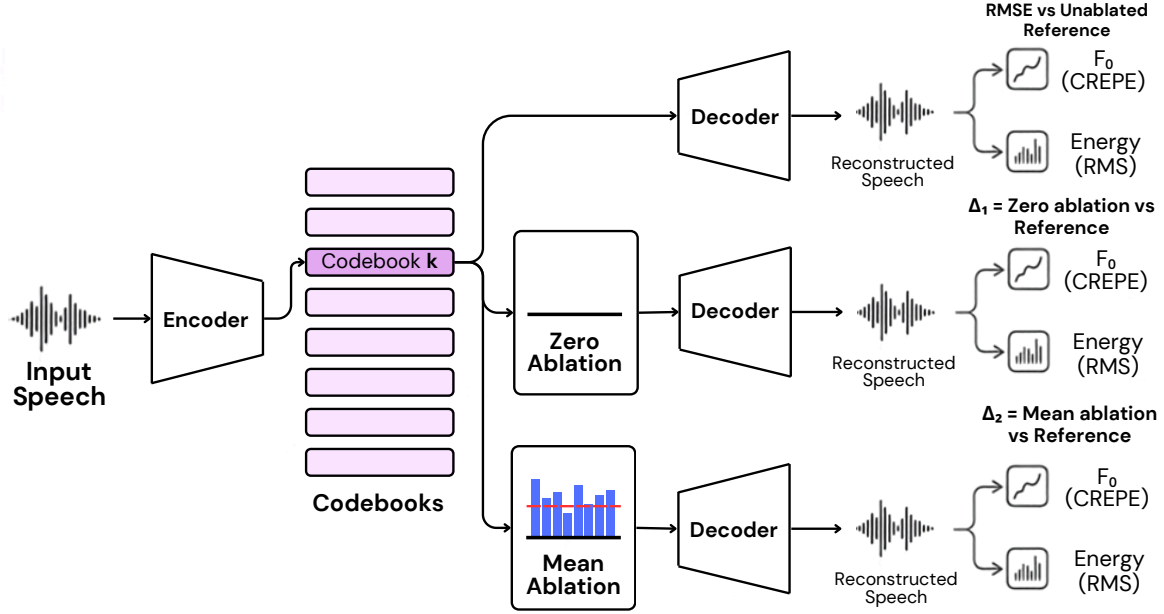


Figure 1: Method schematic. The input speech is encoded into eight RVQ codebooks; a target codebook k is then either left untouched (top, reference), zeroed out (middle, zero-ablation), or replaced with its most frequent code (bottom, mean-ablation), and each of the three resulting code sets is decoded independently. F_0 (CREPE) and energy (RMS) are extracted from each of the three reconstructions, and Δ_1/Δ_2 denote the RMSE of the zero-/mean-ablated reconstruction against the unablated reference, respectively.

Mimi (Défossez et al., 2024). These encourage early codebooks to represent semantic content via self-supervised distillation (Hsu et al., 2021), an architecture underlying speech-language models such as AudioLM (Borsos et al., 2023a) and VALL-E (Wang et al., 2023), but it remains unclear where prosody sits within this hierarchy.

Correlational probing of prosody. Vigneaux and Cole (2026) probe Mimi’s eight codebooks and find pitch distributed across all of them, disputing a clean semantic/acoustic split. Ballier et al. (2026) similarly find, via canonical correlation analysis, that prosody is not cleanly localized to any single codebook. Both share the limitation Elazar et al. (2021) identifies generally: probe accuracy on frozen embeddings shows a codebook is decodable for pitch, not that the decoder causally depends on it.

Robustness benchmarks. Tseng and Harwath (2025) benchmark codecs as black boxes under noise and reverberation (WER, EER, emotion-recognition degradation) across Encodec, DAC (Kumar et al., 2023), SpeechTokenizer, HiFi-Codec (Yang et al., 2023), and FreqCodec (Du et al., 2024), excluding Mimi; Li et al. (2024) similarly

evaluate codecs for SLM-based generation. Neither attributes downstream degradation to a specific internal codebook.

Association models. Sadok et al. (2025) train a bidirectional attribute-token association model mapping RVQ layers of BigCodec, DAC, and SpeechTokenizer to content, speaker, and pitch – richer than a linear probe, but still fit on frozen tokens, not a causal intervention, and not evaluated under noise. Mousavi et al. (2025) survey discrete audio tokenization broadly.

No prior work ablates an RVQ layer, re-decodes, and measures the resulting shift in reconstructed pitch and energy, nor compares that picture across clean and noise-shifted input.

3 Methodology

Experiments are conducted on Mimi (Défossez et al., 2024), which encodes 24 kHz mono audio at 12.5 Hz. The released checkpoint supports up to 32 residual codebooks; we truncate to the first 8 (one semantic codebook distilled from WavLM (Chen et al., 2022), followed by seven cascaded acoustic RVQ codebooks), matching the deployed configuration used by Vigneaux and Cole (2026). Mimi

is trained with quantizer dropout ($\sim 50\%$ probability per sequence), making it robust to missing codebooks and motivating our ablation study.

The transformers implementation of Mimi exposes `audio_codes` of shape (batch, codebooks, time); `decode()` sums the semantic and acoustic contributions. We ablate a codebook by editing its codes before decoding, in two modes: **zero-ablation** (set the codebook’s codes to 0) and **mean-ablation** (replace with the codebook’s globally most frequent code). Mean-ablation exists as a control for a residual-chain confound: zeroing a middle RVQ layer perturbs the residual assumption for every downstream layer, so a gentler perturbation isolates the marginal information content of that layer alone.

For dataset, we use LibriSpeech test-clean (Panayotov et al., 2015) throughout: a preliminary sample of 20 utterances establishes per-codebook baselines, and the main results - F_0 , energy, and STOI, evaluated for every codebook and every utterance come from 200 utterances at three SNR levels (15 dB, 5 dB, 0 dB), with noise injected from MUSAN (Snyder et al., 2015).

Second codec. To assess generality beyond Mimi, we replicate the same protocol on SpeechTokenizer (Zhang et al., 2024), using its standard 8-codebook configuration (one HuBERT-distilled semantic and seven acoustic RVQ codebooks). We retain the same ablation protocol and evaluation metrics, evaluating at 15 dB and 5 dB SNR.

Third codec. To test whether our findings depend on a semantic/acoustic split at all, we further replicate the protocol on EnCodec (Défossez et al., 2022), operated at its 6 kbps setting to match the 8-codebook count of the other two codecs. Unlike Mimi and SpeechTokenizer, EnCodec has no distilled semantic codebook: every codebook is a plain cascaded RVQ layer trained end-to-end, with no self-supervised teacher signal on any one of them. EnCodec shares Mimi’s 24 kHz sample rate, so we reuse the same data preparation and evaluate at all three SNR levels (15 dB, 5 dB, 0 dB).

Metrics. F_0 is extracted with CREPE (Convolutional Representation for Pitch Estimation) (Kim et al., 2018), energy with frame-wise RMS, and intelligibility with the classic variant of STOI (Short-Time Objective Intelligibility) (Taal et al., 2011) – exact hop sizes, voicing threshold, and MUSAN partition are given in Appendix Table 15. All three

metrics are computed from the *output waveform*, which is what makes the measurement causal and output-side rather than a probe on frozen embeddings, and are computed at each codec’s own native sample rate with no cross-rate resampling. F_0 RMSE is computed only over frames voiced in both contours; energy RMSE uses the full frame sequence, trimmed to the shorter reconstruction. The reported effect size is the RMSE between the full (unablated) reconstruction and the layer-ablated reconstruction of *the same condition* - i.e., at each SNR the reference is that SNR’s own unablated reconstruction, not the clean original - so the measurement isolates which codebook is load-bearing under that condition from the fact that noise degrades everything. Effect size is reported in cents for F_0 and dB for energy.

Statistical Analysis. We use a marginal-preserving permutation test (10,000 permutations), permuting noisy-condition per-utterance argmax labels while keeping clean labels fixed, and report bootstrap 95% confidence intervals (10,000 resamples). The seed is fixed for reproducibility, and every test was specified before we inspected the noisy-condition results. Argmax-based tests require a non-NaN F_0 effect for every codebook of a given utterance - CREPE’s voiced-frame threshold occasionally fails to extract a usable contour - so we drop the small number of utterances this affects (12/200 for Mimi, 67/200 for SpeechTokenizer, 55/200 for EnCodec) rather than let an undefined comparison silently corrupt an argmax, and report the exact resulting n throughout.

Confound threshold. A codebook is classified as reconstruction-critical (“confounded”) if its zero-ablation STOI, averaged over the main $n = 200$ sample, falls below 0.85; the rest are treated as intelligibility-preserving and make up the restricted set each stability claim is computed over. This cutoff is a judgment call, not a natural constant, so Section 4.2 reports how the restricted-set ranking changes under two alternative cutoffs for every codec.

4 Results

The results reveal two distinct rankings: one for overall reconstruction, and one for prosody among the codebooks that ranking sets aside – a hierarchy that remains stable under noise and is reproduced across codecs.

Setting	Value
Codecs	Mimi; SpeechTokenizer; EnCodec
Preliminary / main n	20 / 200 utterances
Noise source	MUSAN
SNR levels (Mimi, EnCodec)	15, 5, 0 dB
SNR levels (SpeechTokenizer)	15, 5 dB
Ablation modes	zero, mean
F_0 extractor	CREPE (full)
Energy metric	frame-wise RMS
Permutation N	10,000
Bootstrap N	10,000

Table 1: Experimental setup summary.

4.1 Necessity without specificity

Every codebook’s removal measurably degrades intelligibility, but one codebook breaks it completely. Masking codebook 1 collapses STOI to roughly two-thirds of the lowest value any other codebook reaches (Table 2, full breakdown in Appendix Table 13) - identifying it as uniquely critical for reconstruction. Codebook 1 also shows the largest nominal shift in reconstructed pitch of any codebook in the clean condition, but this effect is confounded by the accompanying collapse in reconstruction quality. A codebook-level correlation between intelligibility damage and pitch-effect size across all eight codebooks is in fact strongly negative, which indicates that at least part of the raw pitch-effect ranking is simply tracking generic collapse rather than anything prosody-specific. We therefore exclude codebook 1 from the prosody analysis and rank only the remaining codebooks, whose ablation preserves intelligibility.

Two codebooks with statistically indistinguishable intelligibility damage produce pitch effects that differ by more than a factor of two (Figure 2; Appendix Figure 6) - a dissociation that would not exist if the pitch-effect ranking were simply reading off how much each codebook’s removal breaks the recording in general.

To verify that the observed confound is not an artifact of this subset, we compute intelligibility for all 200 evaluation utterances and analyze utterance-codebook pairs. Intelligibility and pitch effect remain strongly correlated in the clean condition, and the correlation persists after removing each utterance’s mean intelligibility and pitch effect, indicating that the confound exists within individual utterances rather than arising from differences in utterance difficulty. We therefore treat codebook as the primary unit of analysis rather than attempting to remove the confound statistically. Full intelligi-

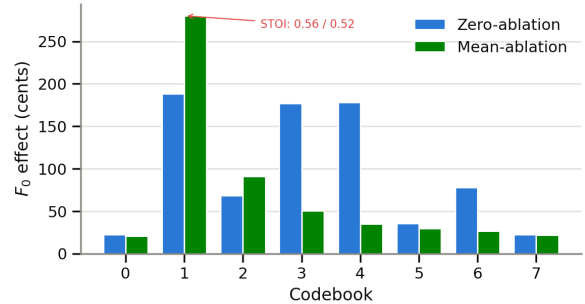


Figure 2: Clean-condition F_0 effect per codebook, zero- vs. mean-ablation. Codebook 1’s STOI is annotated - it is the outlier on intelligibility, not just on nominal F_0 effect.

bility results are reported in Table 2.

4.2 Stability among the rest

Once the confounded codebook is set aside, the remaining seven codebooks produce a ranking of pitch effect (Table 3) that is not just stable across noise conditions but *more* stable than the unfiltered eight-codebook ranking (Table 4, Figure 3). Codebook 1 has the largest mean-ablation effect in every condition (Table 3), consistent with reading it as a general-damage confound rather than a prosody-specific signal; under zero-ablation it is overtaken by codebook 4 at 5 dB and 0 dB. The largest zero-vs-mean disagreement in the grid, however, belongs to codebook 4, not codebook 1: their absolute gap is over four times larger at 5 dB, and by ratio codebook 4’s mean-ablation effect is $5.1 \times$ its zero-ablation effect in the clean condition, against codebook 1’s $1.5 \times$.

For each utterance, the codebook with the largest pitch effect is the same in the clean and noisy conditions more often than expected by chance at every tested noise level, decisively so at 15 dB and 5 dB.

The permutation test is one-sided, testing whether clean and noisy conditions match more

Codebook	Clean		15 dB		5 dB		0 dB	
	Zero	Mean	Zero	Mean	Zero	Mean	Zero	Mean
0	0.963	0.966	0.950	0.950	0.935	0.934	0.926	0.930
1	0.578	0.524	0.531	0.503	0.531	0.521	0.537	0.539
2	0.906	0.897	0.888	0.873	0.862	0.842	0.849	0.833
3	0.906	0.935	0.881	0.915	0.848	0.888	0.825	0.876
4	0.903	0.953	0.882	0.935	0.851	0.910	0.831	0.894
5	0.944	0.963	0.927	0.948	0.905	0.929	0.890	0.911
6	0.947	0.971	0.928	0.955	0.902	0.937	0.887	0.919
7	0.964	0.974	0.952	0.960	0.934	0.941	0.921	0.927

Table 2: Mimi: per-codebook STOI, zero- and mean-ablation, all four conditions, now measured on all $n = 200$ main utterances (not just the $n = 20$ gate sample). Bold row is the STOI-confounded codebook.

Codebook	Clean		15 dB		5 dB		0 dB	
	Zero	Mean	Zero	Mean	Zero	Mean	Zero	Mean
0	22.4	20.6	24.9	25.6	25.8	38.5	32.2	55.8
1	188.4	280.1	193.7	236.5	200.1	242.7	208.7	239.1
2	68.3	90.9	63.9	75.0	58.7	103.0	70.1	98.5
3	177.1	50.7	197.3	48.3	227.3	89.2	254.8	82.0
4	178.1	34.9	195.5	43.6	256.0	59.3	255.4	73.2
5	35.4	29.7	37.3	40.8	50.5	39.7	65.1	64.3
6	77.7	26.4	79.3	33.4	88.0	41.7	100.8	59.9
7	22.5	21.5	25.1	26.8	29.3	40.1	41.6	46.8

Table 3: Mimi: per-codebook F_0 effect (cents), zero- and mean-ablation, all four conditions ($n = 200$). Bold row is the STOI-confounded codebook (§4.1).

often than expected by chance. The observed match rate clears the null decisively at 15 dB and 5 dB ($p < 10^{-3}$), and only marginally at 0 dB ($p = 0.017$, the weakest of the three), consistent with stability degrading gradually rather than holding uniformly at every noise level. The wide bootstrap CIs on ρ indicate that rank correlations are sensitive to small perturbations in the restricted seven-codebook setting. In particular, codebooks 3 and 4 differ by only 1-2 cents in clean and 15 dB, making occasional rank reversals between the zero- and mean-ablation analyses expected.

Near-perfect *aggregate* rank agreement contrasts with a per-utterance argmax match rate only slightly above chance (0.44–0.58 versus a null of 0.38). This dissociation indicates that while the *average* pitch-contribution hierarchy is stable at the population level, the codebook contributing most to pitch differs across individual utterances.

Sensitivity to the confound threshold. The 0.85 STOI cutoff is a judgment call, so we recompute the restricted-set ρ at 0.80 and 0.90 for all three codecs (Appendix Table 11). Mimi is unchanged at all three cutoffs; SpeechTokenizer stays at $\rho = 1.000$ even as the restricted set shrinks from six codebooks to three; EnCodec’s ρ if anything improves

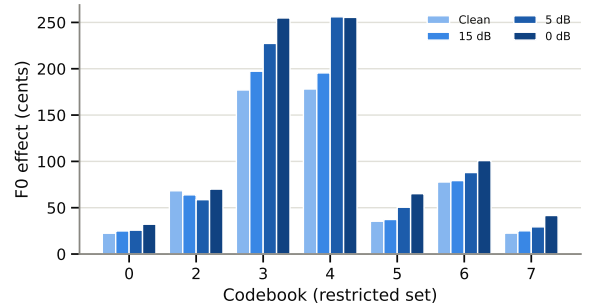


Figure 3: Per-codebook F_0 effect across clean, 15 dB, 5 dB, and 0 dB conditions, restricted to the seven intelligibility-preserving codebooks.

as the cutoff tightens. No cutoff in this range breaks the stability finding.

4.3 Ruling out a trivial explanation

A competing explanation is that the observed stability reflects the marginal frequency with which particular codebooks dominate, rather than utterance-specific effects. If so, agreement should be independent of the margin between the top and second-ranked codebooks in the clean condition. Instead, agreement is consistently higher for utterances with larger margins than for those with smaller margins, across all tested noise levels, with the largest dif-

Cond.	ρ (zero)	ρ (mean)	Match rate (zero)	Null mean / p
15 dB	0.964 [0.929, 1.000]	1.000 [0.786, 1.000]	0.580	0.378 / $< 10^{-4}$
5 dB	1.000 [0.893, 1.000]	0.893 [0.607, 1.000]	0.490	0.379 / 0.0003
0 dB	1.000 [0.893, 1.000]	0.964 [0.607, 1.000]	0.440	0.379 / 0.0170

Table 4: Mimi restricted-set (STOI-stratified, $\{0, 2, 3, 4, 5, 6, 7\}$) stability, $n = 188$ after dropping utterances with an undefined F_0 effect: Spearman ρ between clean and noisy per-codebook F_0 -effect profiles, both ablation modes, with bootstrap 95% CIs (5,000 resamples over utterances); per-utterance argmax match rate, computed under zero-ablation, against its own permutation null.

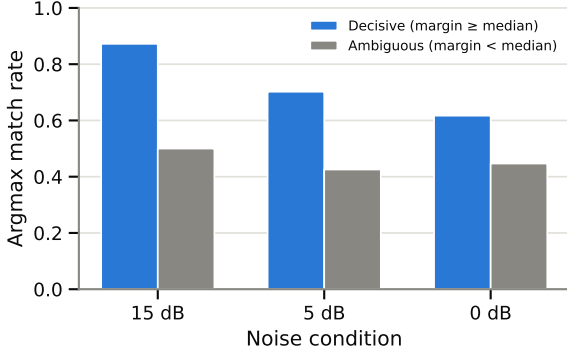


Figure 4: Argmax match rate for decisive vs. ambiguous utterances (split at the median clean-condition top-1/top-2 margin), across all three SNR levels.

ference at 15 dB. This pattern is inconsistent with a purely marginal-distribution explanation.

4.4 Generalization across codecs

To assess generality, we repeat the analysis on SpeechTokenizer, which uses a different architecture despite the same split-RVQ formulation and 8-codebook configuration. We observe that the codebooks with the largest raw pitch effects also exhibit the largest intelligibility degradation. Unlike Mimi, however, the confound is distributed across four reconstruction-critical codebooks rather than a single dominant one (Tables 5 and 6).

Restricting the analysis to the remaining non-confounded codebooks, the ranking is highly stable: rank correlation between clean and noisy pitch-effect profiles is near-perfect at both tested noise levels, and per-utterance agreement clears its own permutation baseline by a wide margin (Appendix Table 9, Figure 5). As with Mimi, zero and mean-ablation diverge on the aggregate winner under noise – here even in sign, not just magnitude – which is why we continue to rely on each mode’s own cross-condition consistency rather than a single pooled number.

EnCodec offers a sharper test of generality, since it has no semantic codebook at all: every one of

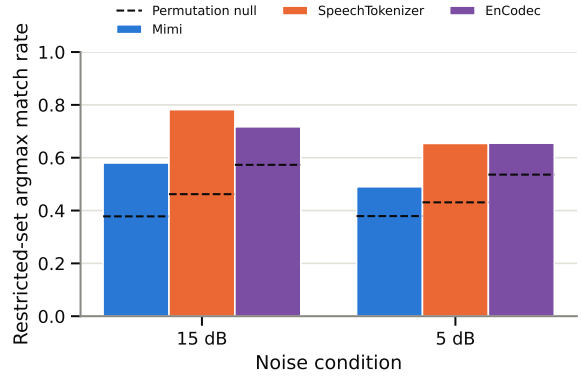


Figure 5: Restricted-set argmax match rate, Mimi vs. SpeechTokenizer vs. EnCodec, shared 15/5 dB conditions. Dashed lines mark each codec’s own permutation null.

its eight codebooks is a plain, cascaded RVQ layer with no self-supervised distillation target. Despite that, the confound reappears in the same shape it took on Mimi, not the diffuse form it took on SpeechTokenizer – a single codebook (codebook 0) collapses intelligibility while the rest stay well above the confound threshold (Tables 7 and 8). One codebook among the remaining seven shows a raw pitch effect more than twice as large as any other, without a matching drop in intelligibility, the same necessity-without-specificity pattern documented for Mimi in Section 4.1 (Appendix Figure 8).

Restricted to the seven intelligibility-preserving codebooks, the ranking is stable across all three noise levels, clearing its own permutation baseline by a wide margin at every one (Appendix Table 10, Figure 5). Zero- and mean-ablation agree at 15 dB and 5 dB and diverge only at 0 dB, the mildest disagreement of the three codecs – consistent with the pattern that this disagreement is a property of each codec’s own residual chain rather than of the ablation procedure itself.

Codebook	Clean		15 dB		5 dB	
	Zero	Mean	Zero	Mean	Zero	Mean
0	28.8	29.9	26.6	55.6	22.3	48.3
1	49.3	90.6	47.8	48.0	50.8	60.5
2	29.0	27.5	29.9	38.7	38.3	48.0
3	20.7	22.9	21.9	24.4	23.9	32.9
4	339.8	20.4	349.9	24.4	351.9	26.0
5	280.5	17.3	318.3	18.0	347.4	25.6
6	104.4	16.5	83.9	19.5	158.0	25.9
7	110.8	16.4	143.2	17.0	215.4	20.4

Table 5: SpeechTokenizer: per-codebook F_0 effect (cents), zero- and mean-ablation, all three conditions ($n = 200$). Bold rows are the four STOI-confounded codebooks (§4.4) – a more diffuse confound structure than Mimi’s single codebook.

Codebook	Clean		15 dB		5 dB	
	Zero	Mean	Zero	Mean	Zero	Mean
0	0.651	0.566	0.543	0.471	0.482	0.451
1	0.864	0.864	0.861	0.853	0.838	0.830
2	0.905	0.921	0.880	0.899	0.845	0.867
3	0.956	0.940	0.940	0.924	0.917	0.897
4	0.710	0.960	0.662	0.940	0.614	0.912
5	0.929	0.969	0.905	0.952	0.875	0.934
6	0.809	0.973	0.761	0.957	0.713	0.937
7	0.825	0.978	0.783	0.961	0.737	0.940

Table 6: SpeechTokenizer: per-codebook STOI, zero- and mean-ablation, all three conditions, measured on all $n = 200$ main utterances. Bold rows are the four STOI-confounded codebooks.

5 Discussion

The overall pattern holds regardless of the specific codebooks involved: a small subset is critical for reconstruction, and the rest exhibit a stable pitch hierarchy, on Mimi and SpeechTokenizer (both semantically distilled) and on EnCodec (no semantic codebook at all). This suggests the confound is a property of cascaded residual quantization itself, not of how any codebook was trained – so the finding is structural, not codebook-specific: pruning should preferentially retain codebooks ranked high in the restricted-set hierarchy, a prediction we motivate but do not test here.

Zero and mean-ablation agree only partially even in clean conditions ($\rho \approx 0.68$ on Mimi) and diverge further under noise, most severely on SpeechTokenizer (negatively correlated). This disagreement is present from the start, which we attribute to zero-ablation’s harsher disruption of the residual chain; each mode stays internally consistent across noise levels, so our conclusions rest on within-method rather than cross-method agreement.

Pitch and energy also diverge under noise: the restricted-set F_0 effect is roughly flat, while energy shrinks steadily for every codebook (Appendix

Figure 10), consistent with rising background energy compressing the headroom for any ablation to move it – We therefore treat energy as a secondary, confirmatory measure rather than an independent finding. Finally, the dose-response analysis (Figure 4) argues against a purely aggregate-effect explanation: agreement increases with the clean-condition top-two margin, a pattern not expected under a fixed marginal-distribution artifact.

6 Conclusion

Using causal output-side ablation rather than probing, we show that the causal organization of prosody in neural audio codecs is best understood in two tiers. A codec-specific subset of codebooks is reconstruction-critical and confounded with intelligibility, whereas the remaining intelligibility-preserving codebooks exhibit a stable causal hierarchy for pitch under additive noise. This two-tier structure generalizes across Mimi, SpeechTokenizer, and EnCodec despite their architectural differences – including the presence or absence of a distilled semantic codebook – although the reconstruction-critical codebooks themselves are codec-dependent.

Codebook	Clean		15 dB		5 dB		0 dB	
	Zero	Mean	Zero	Mean	Zero	Mean	Zero	Mean
0	960.6	38.4	869.4	42.3	786.8	55.1	732.6	56.7
1	64.5	24.0	121.1	27.3	190.9	48.4	204.2	59.1
2	352.9	20.4	266.6	21.9	263.4	31.4	274.0	38.8
3	800.4	20.2	781.3	18.5	780.1	25.8	753.4	21.8
4	214.9	17.1	121.2	16.3	110.1	25.3	91.9	19.4
5	167.7	15.2	54.7	17.2	49.3	14.9	39.5	21.5
6	14.6	15.9	15.9	14.2	18.5	15.2	16.8	17.0
7	41.6	15.4	24.7	15.3	21.4	16.3	22.4	15.9

Table 7: EnCodec: per-codebook F_0 effect (cents), zero- and mean-ablation, all four conditions ($n = 200$). Bold row is the STOI-confounded codebook (Section 4.4) – EnCodec has no semantic/acoustic split, yet the confound still lands on a single codebook, as with Mimi.

Codebook	Clean		15 dB		5 dB		0 dB	
	Zero	Mean	Zero	Mean	Zero	Mean	Zero	Mean
0	0.607	0.661	0.558	0.626	0.548	0.623	0.554	0.627
1	0.876	0.903	0.865	0.888	0.849	0.869	0.840	0.859
2	0.897	0.945	0.882	0.932	0.862	0.917	0.852	0.909
3	0.935	0.959	0.922	0.947	0.906	0.934	0.896	0.926
4	0.948	0.969	0.936	0.960	0.923	0.948	0.915	0.942
5	0.946	0.975	0.933	0.967	0.920	0.958	0.911	0.951
6	0.969	0.980	0.960	0.972	0.952	0.962	0.947	0.956
7	0.971	0.983	0.961	0.976	0.951	0.968	0.944	0.963

Table 8: EnCodec: per-codebook STOI, zero- and mean-ablation, all four conditions, measured on all $n = 200$ main utterances. Bold row is the STOI-confounded codebook.

7 Limitations

Coverage. All three codecs are cascaded RVQ designs on read speech (LibriSpeech); codecs built on different quantization schemes, multilingual and expressive/spontaneous speech, and noise types beyond additive stationary noise remain untested.

The interventions themselves are imperfect. Zeroing or mean-replacing a middle codebook puts the decoder in a residual state its training never produced (quantizer dropout and variable-bandwidth training both drop chain *suffixes*, not interior layers), so our effect size conflates causal reliance with brittleness to an off-manifold residual; *resample-ablation* (swap codebook k ’s codes with another utterance’s) is a cleaner future control. Relatedly, the confound classification rests entirely on STOI, a known-weak proxy under a codec decoder’s non-linear distortions – an ASR-WER column, as Tseng and Harwath (2025) use, would harden it. And zero/mean-ablation disagreement is itself architecture-dependent (mild on Mimi/EnCodec, sharp and sign-flipping on SpeechTokenizer) for reasons we do not fully resolve.

We motivate but do not test the probe/ablation dissociation ourselves. The paper’s premise rests on prior probing work (Elazar et al., 2021; Vigneaux and Cole, 2026; Ballier et al., 2026); fitting our own linear pitch probe on the same 200 utterances and comparing it against our ablation ranking directly is the natural next step, not something this version does.

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A Additional Figures

The figures collected here provide the complete evidence underlying the main-text claims; each one is the evidence a specific sentence in Section 4.1 or Section 4.4 depends on, expanded to the full

codebook grid or the full utterance set rather than the summary numbers quoted inline.

Figure ?? illustrates the necessity-without-specificity dissociation argued for in Section 4.1 by plotting intelligibility damage against raw pitch effect for every codebook at once. Figure 6 does this for Mimi: codebook 1 sits alone in the lower-left, decisively worse on STOI than anything else, while codebooks 2, 3, and 4 cluster at nearly identical STOI yet spread across more than a factor of 2.5 in pitch effect – exactly the pattern that rules out “STOI damage explains the pitch ranking” as a complete account. Figure 7 repeats this for SpeechTokenizer, where the confound is visibly less clean: four points, not one, fall into the lower-left region, and the codebook with the single largest pitch effect is one of them. Figure 8 repeats it again for EnCodec, whose codebooks have no distillation target of any kind; the shape here matches Mimi’s, a single isolated outlier, rather than SpeechTokenizer’s more diffuse one, and one of the seven remaining codebooks (codebook 3) shows a pitch effect nearly as large as the confounded codebook’s without any corresponding drop in intelligibility – the sharpest individual dissociation observed across all three codecs.

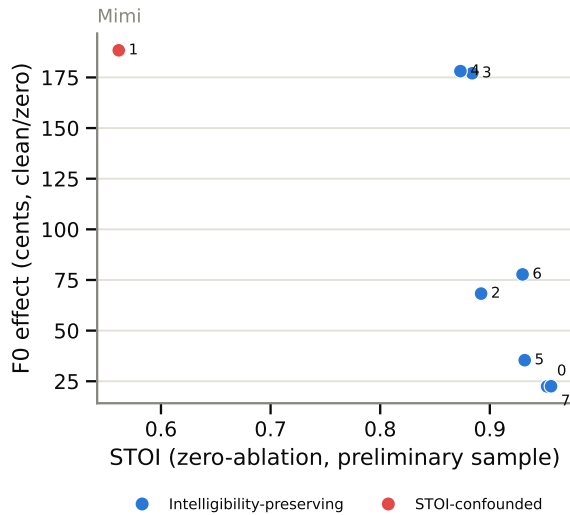


Figure 6: Mimi: per-codebook STOI (zero-ablation, preliminary $n = 20$ sample) vs. clean-condition F_0 effect (zero-ablation, $n = 200$), all 8 codebooks. Codebooks 2, 3, and 4 show near-identical STOI but a more than 2.5-times spread in F_0 effect.

The ablation-mode complication discussed in the Discussion section is likewise easier to judge from the full scatter than from a summary correlation. Points that fall on the dotted diagonal in Figure 9

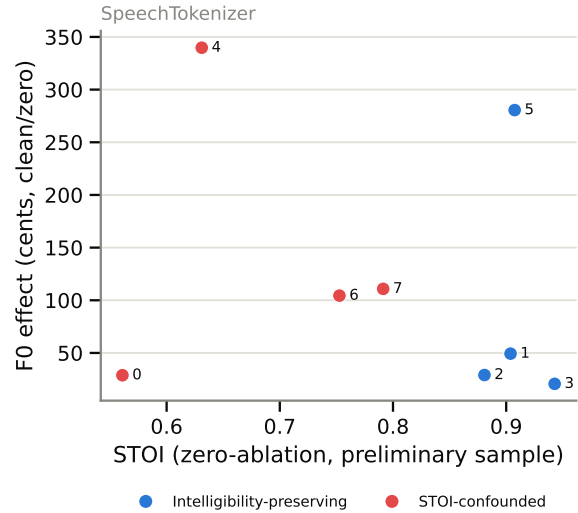


Figure 7: SpeechTokenizer: the same diagnostic, all 8 codebooks. Four codebooks (red) fall below the confound threshold rather than one, and the codebook with the largest F_0 effect (4) is itself confounded - a more diffuse failure shape than Mimi’s.

are conditions where zero- and mean-ablation agree on a codebook’s pitch effect; codebook 1 sits far off that diagonal in every panel, and the whole-sample scatter drifts further away as noise increases, which is the visual form of the divergence reported numerically in the main text.

Finally, Figure 10 gives the energy-channel counterpart of the main F_0 stability result. Energy effects behave differently from pitch under noise: rather than the roughly flat or non-monotonic trajectory F_0 effect shows across conditions, energy effects shrink steadily as noise increases for every codebook, consistent with rising background energy compressing the headroom left for any one codebook’s ablation to move the signal.

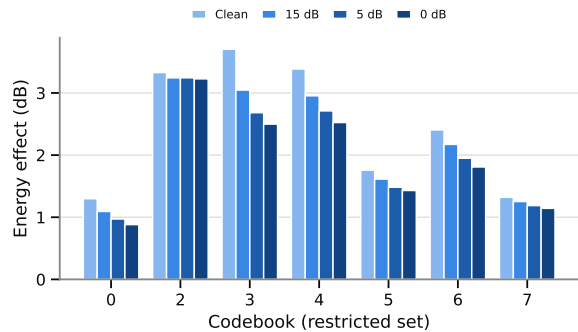


Figure 10: Mimi: per-codebook energy effect (dB), zero-ablation, restricted set, all four conditions - the energy-channel counterpart of Figure 3. Energy effects shrink with more noise for every codebook, unlike F_0 effect’s non-monotonic pattern.

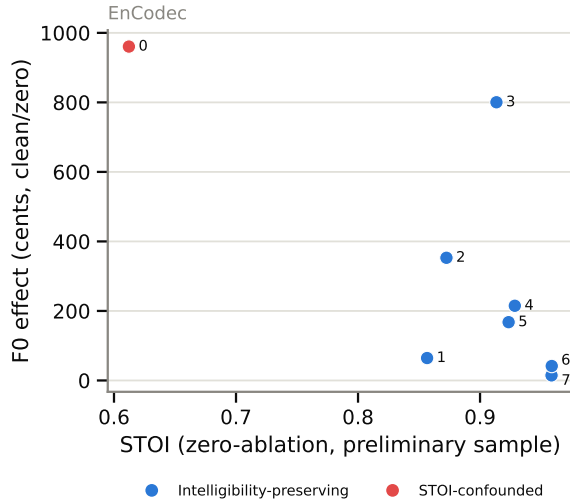


Figure 8: EnCodec: the same diagnostic, all 8 codebooks. A single codebook (0) is confounded, as with Mimi, despite EnCodec having no semantic codebook at all; codebook 3 shows the largest and cleanest necessity-without-specificity gap of any codebook in any of the three codecs.

B Additional Tables

The tables below provide the full numerical results underlying the main-text summaries: the SpeechTokenizer and EnCodec restricted-set stability tables referenced in Section 4.4, the confound-threshold sensitivity check referenced in Section 4.2, per-codebook STOI on the preliminary sample for all three codecs, the permutation-null detail behind Figure 5, the hyperparameters held fixed throughout, and the full energy-effect grids that Section 4.1 onward only characterizes in aggregate.

Table 12 reports the restricted-set match rate, its permutation null, and the resulting one-sided p -value for all three codecs at their shared 15/5,dB conditions.

Codec / cond.	Match rate	Null mean	p
Mimi, 15 dB	0.580	0.378	$< 10^{-4}$
Mimi, 5 dB	0.490	0.379	0.0003
SpeechTokenizer, 15 dB	0.782	0.462	$< 10^{-4}$
SpeechTokenizer, 5 dB	0.654	0.431	$< 10^{-4}$
EnCodec, 15 dB	0.717	0.573	$< 10^{-4}$
EnCodec, 5 dB	0.655	0.536	0.0002

Table 12: Restricted-set argmax match rate against its own permutation null, all three codecs, shared 15/5 dB conditions – the tabulated version of Figure 5.

Table 13 is the preliminary-sample STOI table for Mimi; the corresponding numbers for EnCodec appear in Table 14 and follow the same single-

codebook confound shape.

Codebook	STOI, zero	STOI, mean
0	0.952	0.960
1	0.561	0.516
2	0.893	0.870
3	0.885	0.912
4	0.873	0.938
5	0.931	0.953
6	0.931	0.963
7	0.956	0.967

Table 13: Mimi: per-codebook STOI, zero- and mean-ablation, preliminary $n = 20$ sample.

Codebook	STOI, zero	STOI, mean
0	0.612	0.689
1	0.857	0.883
2	0.872	0.931
3	0.913	0.947
4	0.928	0.959
5	0.923	0.967
6	0.959	0.972
7	0.959	0.977

Table 14: EnCodec: per-codebook STOI, zero- and mean-ablation, preliminary $n = 20$ sample. Codebook 0 falls to 71% of the lowest value any other codebook reaches (0.612/0.857) – a smaller relative gap than Mimi’s 64% (Table 13), despite EnCodec having no distillation target on any codebook, including this one.

Energy effects are reported throughout in the same format as the F_0 tables in the main text, both ablation modes, all conditions, for each of the three codecs in turn.

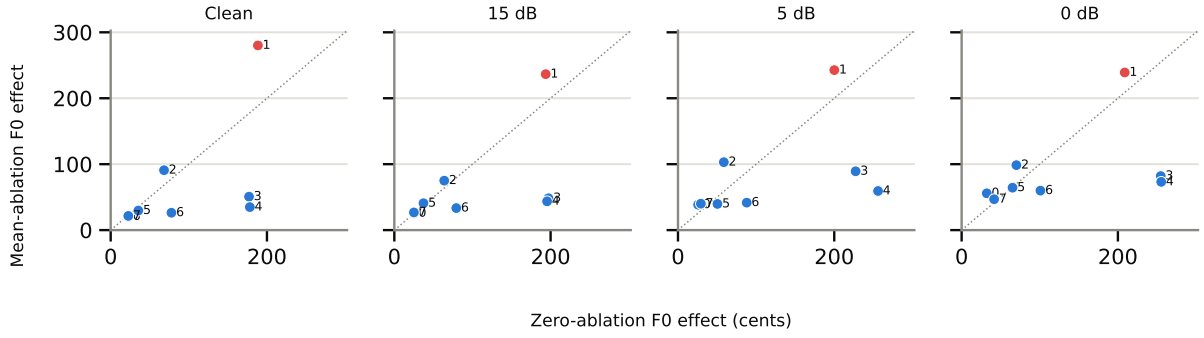


Figure 9: Mimi: zero- vs. mean-ablation F_0 effect per codebook, all four conditions. Points on the dotted diagonal agree between ablation modes; codebook 1 (red) sits far off it in every panel, and the whole-sample scatter drifts further from the diagonal as noise increases.

Cond.	ρ (zero)	ρ (mean)	Match (zero)	Null / p
15 dB	1.000 [1.000, 1.000]	1.000 [0.800, 1.000]	0.782	$0.462 / < 10^{-4}$
5 dB	1.000 [0.800, 1.000]	1.000 [0.800, 1.000]	0.654	$0.431 / < 10^{-4}$

Table 9: SpeechTokenizer restricted-set (STOI-stratified, $\{1, 2, 3, 5\}$) stability, $n = 133$: Spearman ρ (clean vs. noisy), both ablation modes, bootstrap 95% CIs; per-utterance argmax match rate under zero-ablation against its own permutation null.

Cond.	ρ (zero)	ρ (mean)	Match (zero)	Null / p
15 dB	0.964 [0.893, 0.964]	0.750 [0.429, 0.964]	0.717	$0.573 / < 10^{-4}$
5 dB	0.893 [0.857, 0.893]	0.964 [0.571, 1.000]	0.655	$0.536 / 0.0002$
0 dB	0.893 [0.857, 0.893]	0.786 [0.428, 0.964]	0.634	$0.514 / 0.0002$

Table 10: EnCodec restricted-set (STOI-stratified, $\{1, 2, 3, 4, 5, 6, 7\}$) stability, $n = 145$: Spearman ρ (clean vs. noisy), both ablation modes, bootstrap 95% CIs; per-utterance argmax match rate under zero-ablation against its own permutation null.

Codec / threshold	Kept	ρ (15/5/0 dB)
Mimi, 0.80	7	0.964 / 1.000 / 1.000
Mimi, 0.85	7	0.964 / 1.000 / 1.000
Mimi, 0.90	7	0.964 / 1.000 / 1.000
SpeechTokenizer, 0.80	6	1.000 / 1.000 / –
SpeechTokenizer, 0.85	4	1.000 / 1.000 / –
SpeechTokenizer, 0.90	3	1.000 / 1.000 / –
EnCodec, 0.80	7	0.964 / 0.893 / 0.893
EnCodec, 0.85	7	0.964 / 0.893 / 0.893
EnCodec, 0.90	5	1.000 / 1.000 / 1.000

Table 11: Restricted-set Spearman ρ (clean vs. each noisy condition, zero-ablation) under three alternative STOI-confound thresholds. “Kept” is the number of codebooks retained. SpeechTokenizer has no 0 dB condition.

Setting	Value
CREPE model	full, 10 ms hop
Voicing threshold (F_0)	periodicity > 0.5
Energy hop	240 samples (10 ms@24k, 15 ms@16k)
STOI variant	classic (non-extended)
MUSAN partition	noise only
Permutation N	10,000
Bootstrap N	10,000
Seed	fixed across all experiments
Preliminary STOI threshold	0.3
STOI confound threshold	0.85
Preliminary / main n	20 / 200

Table 15: Hyperparameters used throughout.

Codebook	Clean		15 dB		5 dB		0 dB	
	Zero	Mean	Zero	Mean	Zero	Mean	Zero	Mean
0	1.30	1.21	1.09	1.03	0.97	0.92	0.88	0.82
1	12.40	19.88	9.82	14.42	8.58	10.73	8.62	9.23
2	3.33	3.24	3.24	3.30	3.24	3.44	3.23	3.32
3	3.70	2.66	3.05	2.61	2.68	2.43	2.50	2.38
4	3.38	2.11	2.95	1.92	2.71	1.88	2.52	1.77
5	1.76	2.08	1.61	1.92	1.48	1.73	1.43	1.77
6	2.40	1.56	2.17	1.50	1.95	1.51	1.81	1.44
7	1.32	1.45	1.25	1.47	1.19	1.38	1.14	1.35

Table 16: Mimi: per-codebook energy effect (dB), zero- and mean-ablation, all four conditions ($n = 200$).

Codebook	Clean		15 dB		5 dB	
	Zero	Mean	Zero	Mean	Zero	Mean
0	11.57	20.09	8.64	13.63	7.15	10.62
1	4.99	6.40	5.06	5.21	5.26	4.57
2	2.46	3.19	2.32	3.24	2.15	2.85
3	1.53	2.44	1.43	2.20	1.30	1.93
4	8.45	2.26	7.02	1.72	5.88	1.58
5	1.51	1.52	1.40	1.43	1.32	1.26
6	9.96	1.40	8.94	1.38	7.84	1.27
7	6.90	1.16	5.16	1.06	4.18	0.99

Table 17: SpeechTokenizer: per-codebook energy effect (dB), zero- and mean-ablation, all three conditions ($n = 200$).

Codebook	Clean		15 dB		5 dB		0 dB	
	Zero	Mean	Zero	Mean	Zero	Mean	Zero	Mean
0	29.38	17.59	25.53	12.48	22.14	9.76	19.92	9.05
1	4.21	2.51	3.21	2.69	2.82	2.49	2.65	2.55
2	4.18	1.53	2.82	1.54	2.40	1.43	2.21	1.39
3	3.34	1.06	1.92	1.15	1.53	1.13	1.36	1.07
4	2.50	0.92	1.67	1.03	1.38	0.99	1.23	0.95
5	2.24	0.77	1.53	0.75	1.30	0.68	1.17	0.66
6	0.83	0.59	0.71	0.61	0.65	0.63	0.61	0.63
7	1.24	0.63	0.91	0.78	0.77	0.66	0.69	0.64

Table 18: EnCodec: per-codebook energy effect (dB), zero- and mean-ablation, all four conditions ($n = 200$).